Robust Charging Infrastructure Design under Demand Uncertainty

A solar-powered charging station for a shared e-micromobility fleet

Which station do you build when the future is uncertain?

Theme 3 — solar charging station

Theme 2 — charge/discharge profiles

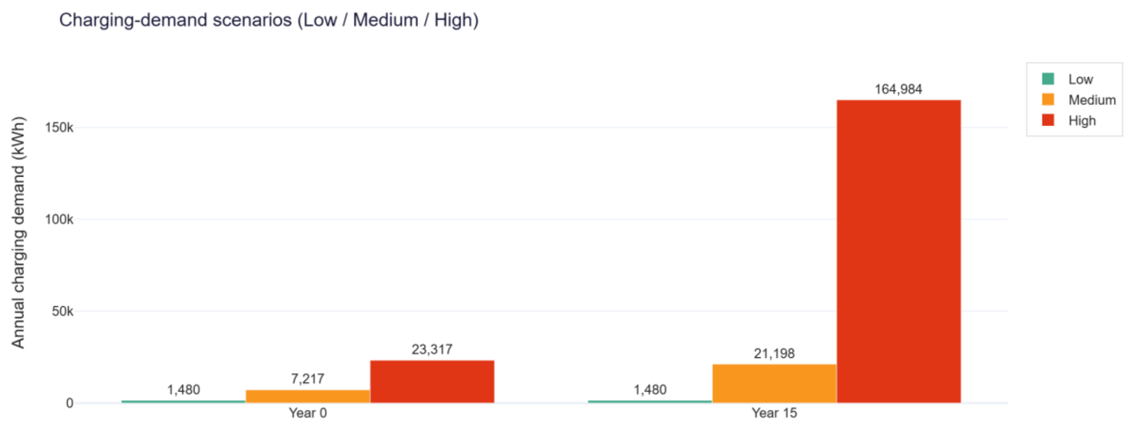
The problem: sized for average, or for the worst?

A mixed-fleet depot. The hub charges e-scooters, e-bikes and e-cargo bikes; demand swings with season, weather, events and growth.

The dilemma. Size for average demand and the fleet is stranded at peaks; size for the worst case and capital is wasted.

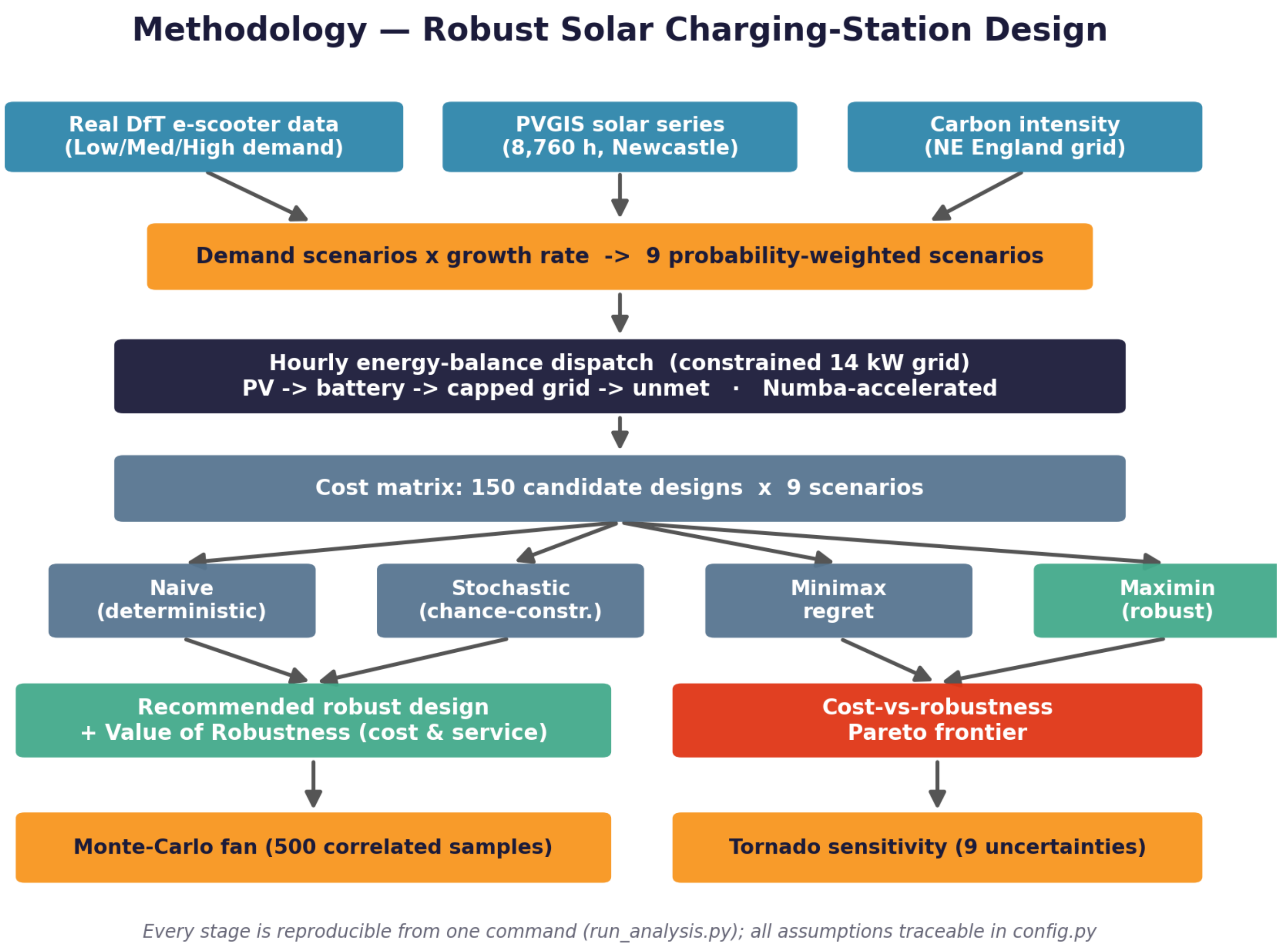
A constrained grid. A higher import limit needs a costly DNO reinforcement, so the evening peak must come from on-site solar + storage.

Real data. Low / Medium / High scenarios are built from the real DfT Newcastle e-scooter trial — trips, fleet, deployment, distance.

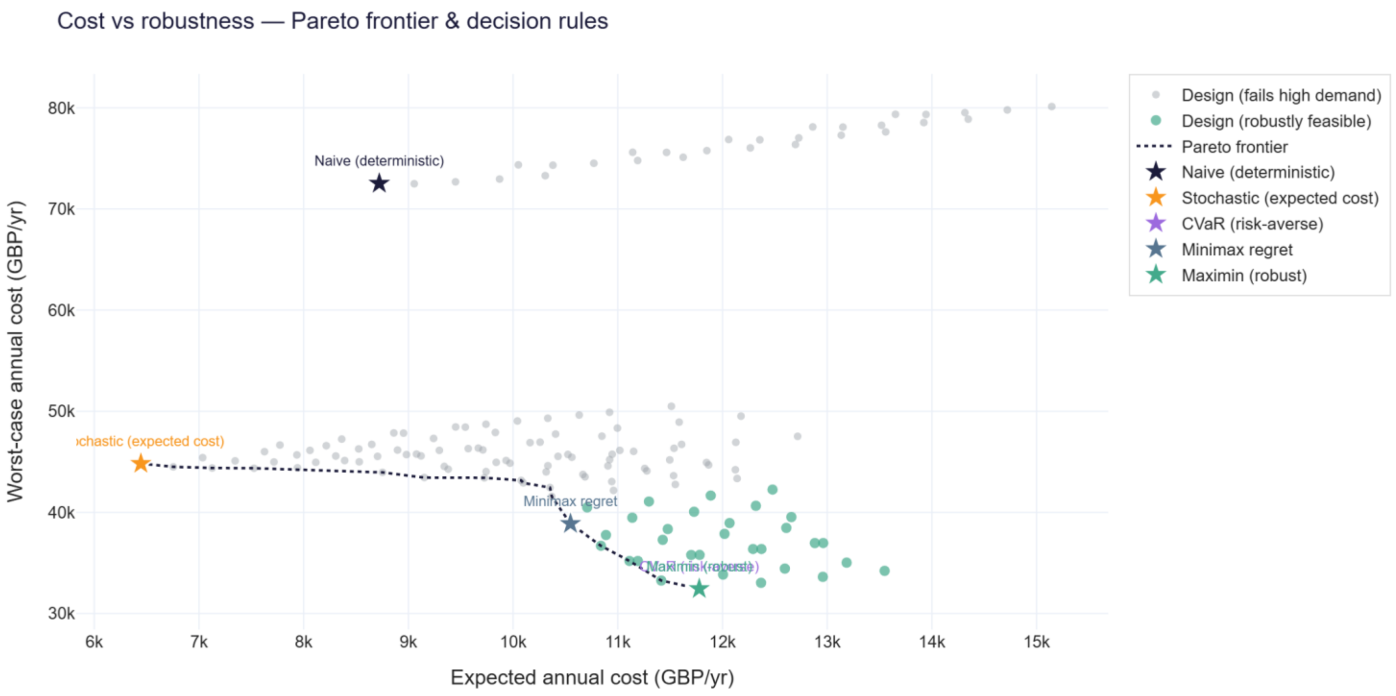


Demand scenarios derived from real trial data.

Approach — a reproducible robust-design pipeline



The result: robustness pays



-55%

worst-case annual cost vs the naive design

87% -> 97%

guaranteed fleet service, worst demand

£32,450/yr

robust worst-case cost (down from £72,523)

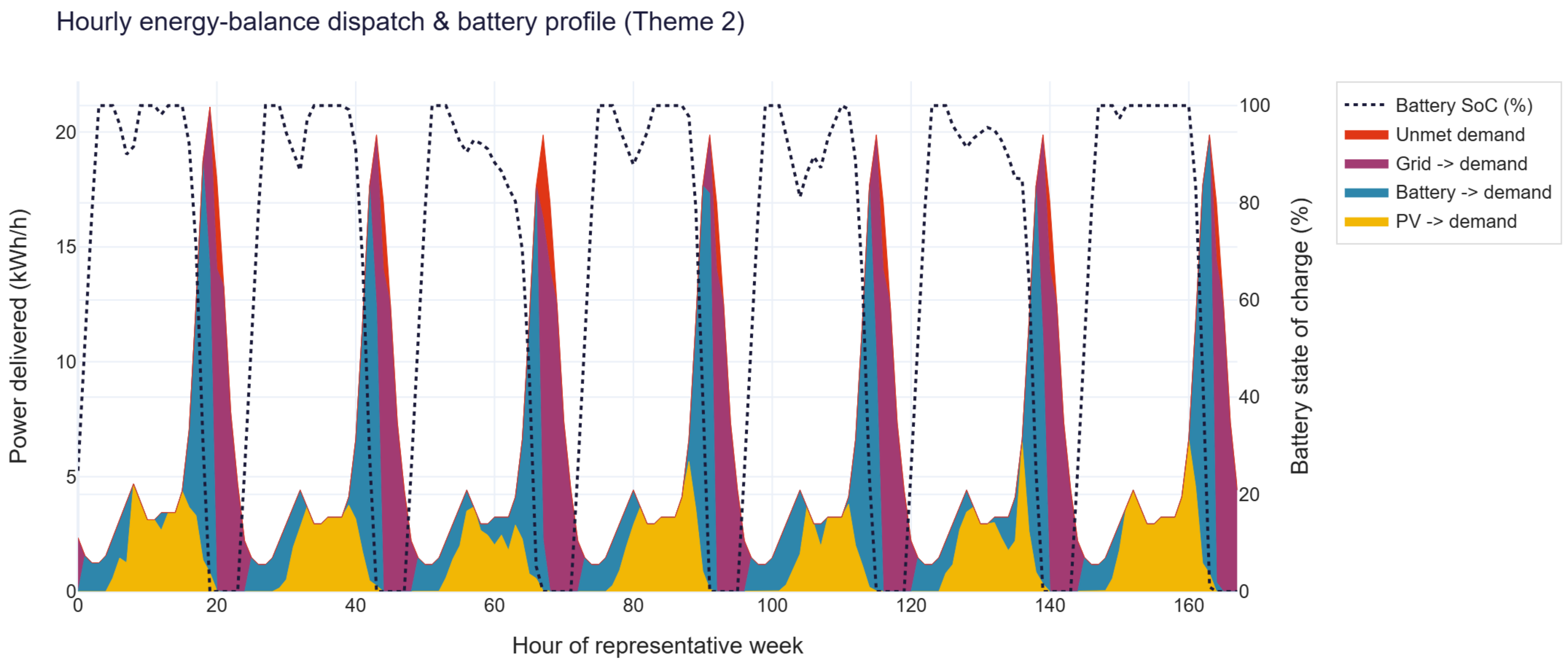
A small expected-cost premium buys large protection against the demand tail.

Recommended robust station — Theme 2 in action

25 kWp solar PV

50 kWh battery

8 charge points



Battery charges overnight and at midday, then discharges through the evening peak the capped grid cannot meet.

Feasible today — and sustainable

~£63,600

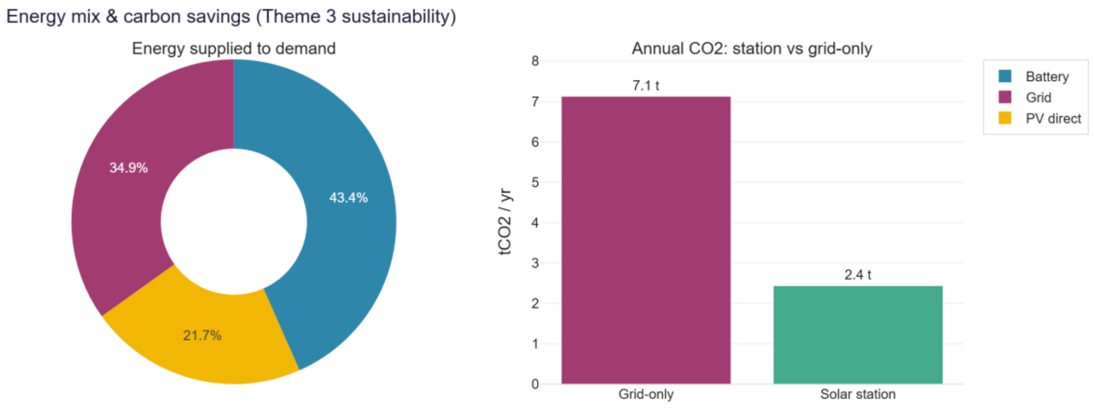
capital cost — off-the-shelf PV, Li-ion & AC chargers

-66% CO_2

vs grid-only charging (4.7 tCO_2/yr saved)

7/7 validated

key outputs within published benchmark ranges; 1-command rebuild



Original, authentic, reproducible

Every line of code and every figure is original, written from scratch. All three datasets are real: DfT e-scooter data (Open Government Licence), live PVGIS solar and National Grid ESO carbon intensity. Fixed seed, one-command rebuild, passing tests — built for the AI and plagiarism checks.

We turn demand uncertainty from a risk
into a design input — and show that robustness pays.

Thank you.